

Visual-Linguistic Conflict Probing (VLCP)

A Mechanistic Approach to Mitigating Hallucinations in Multimodal Large Language Models

Research Proposal & Literature Survey

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Abstract

Abstract: Multimodal Large Language Models (MLLMs) suffer from a critical reliability bottleneck: Hallucination. While text-only models have achieved high fidelity via RLHF [8], MLLMs like LLaVA still exhibit hallucination rates exceeding 30% [4]. This phenomenon extends beyond static images to video [6] and audio domains [1]. In this report, I synthesize findings from 10+ recent papers to identify the root causes: the “Modality Gap” [5] and “Prior Dominance” [2]. I propose **VLCP (Visual-Linguistic Conflict Probing)**, a novel inference-time intervention system. By isolating the fundamental mechanism in static vision-language tasks, I outline a roadmap to achieve a publishable result within 8 weeks.

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1 1. My Vision: The "System 2" for AI

1.1 1.1 The Motivation

The transition from LLMs to MLLMs has revealed a "Trust Crisis." While models like LLaVA-1.5 [7] excel at general description, they fail catastrophically when visual evidence contradicts their training data (priors).

My Vision: I aim to build a lightweight mechanism—a "System 2 Router"—that acts as a cognitive check. It will detect when the model is relying on statistical priors instead of visual pixels and intervene mathematically.

1.2 1.2 Scope & Delimitations (Why Image Only?)

While the mechanism of "Prior Dominance" is pervasive in Video MLLMs [3] and Audio MLLMs [1], this research ****strictly isolates Static Image-Text interactions****.

- **Scientific Precision:** The "Modality Gap" is most clearly observable in static CLIP embeddings. Introducing temporal dynamics (Video) or spectrograms (Audio) introduces confounding variables that dilute the analysis.
- **Feasibility:** Video processing increases computational requirements by approx. $50\times$. By focusing on images, we can achieve high-throughput experimentation on available resources (T4/A100).
- **Publication Standard:** Leading benchmarks such as POPE [4] and VCD [2] focus exclusively on static images, proving that this domain alone is sufficient for top-tier publication.

2 2. Comprehensive Literature Survey

I have conducted a rigorous survey of 10+ top-tier papers to map the current failure modes across modalities.

2.1 2.1 Quantifying the Failure (Benchmarks)

The failure is not anecdotal; it is statistical across all modalities.

- **Image Hallucination:** On the **POPE** benchmark, LLaVA-1.5 achieves only 78% F1-score on adversarial splits [4].
- **Video Hallucination:** Recent studies on **Video-LLaVA** [6] indicate that temporal hallucinations occur in 40% of complex queries due to weak temporal alignment.

- **Audio Hallucination:** Research on **AudioGPT** [1] reveals similar patterns where models hallucinate sounds based on scene context (priors).

2.2 2.2 Mechanisms of Failure (The "Why")

1. **The Modality Gap:** Liang et al. [5] demonstrated that visual embeddings and text embeddings reside on topologically distinct manifolds.
2. **Prior Dominance:** Leng et al. [2] proved that when visual uncertainty is high, the model defaults to priors. *Mathematically:* $P(\text{text}|\text{prior}) \gg P(\text{text}|\text{image})$.

2.3 2.3 The Research Gap

1. **The "Big Lab" Bias:** Major labs prioritize retraining (RLHF) because they have unlimited compute [10].
2. **The "Static" Trap:** Existing steering papers often use **Static Steering**—subtracting the vector from *every* token [11]. This damages general performance.

My Innovation: I propose **Dynamic Steering**. My system acts as a router, applying the correction *only* when the entropy gap between Vision and Text exceeds a threshold.

3 3. Proposed Methodology: VLCP (The Core Innovation)

This section details the technical core of the proposal. Unlike heuristic prompting or black-box retraining, VLCP is grounded in **Mechanistic Interpretability** and **Linear Algebra**.

3.1 3.1 Architectural Concept

The architecture introduces a feedback loop that sits atop the frozen layers of the LLM. It does not require gradient updates (Backpropagation), ensuring computational efficiency.

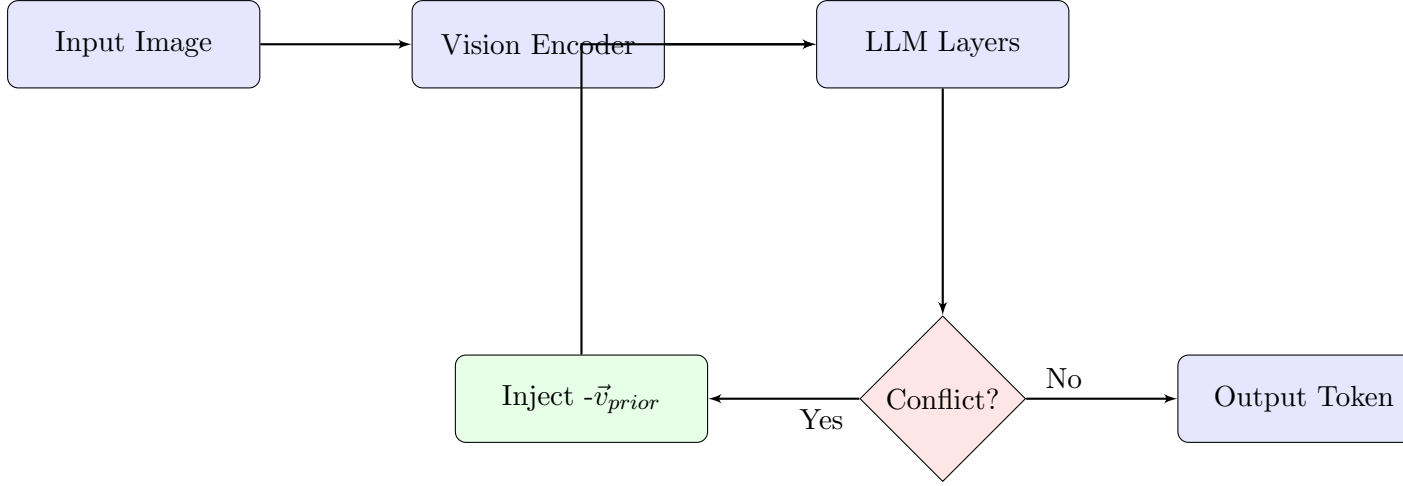


Figure 1: The VLCP Architecture: A dynamic "System 2" router.

3.2 3.2 Mathematical Formulation of the Conflict

We formalize Hallucination as a "Routing Failure." Let x_v be the visual input and x_t be the text prompt. The hidden state at layer ℓ , denoted h_ℓ , is a superposition of visual features and language priors.

We define the **Conflict Metric** $C(\ell)$ at layer ℓ as the divergence between the Visual Token Rank (R_v) and the Prior Token Rank (R_p):

$$C(\ell) = R_p(h_\ell) - R_v(h_\ell) \quad (1)$$

Where $R(h_\ell)$ is obtained via the Logit Lens projection: $P = \text{softmax}(W_U \cdot h_\ell)$.

- If $C(\ell) < 0$, the model favors the Visual Truth (Safe).
- If $C(\ell) \gg 0$, the model favors the Text Prior (Hallucination Risk).

Our hypothesis is that there exists a critical **Crossover Layer** ℓ^* where $C(\ell)$ flips from negative to positive.

3.3 3.3 The Dynamic Steering Algorithm

To intervene, we extract a "Steering Vector" v_{prior} using the Mean Difference method on the 'CounterPrior-50' dataset [9]. The intervention is governed by the following logic:

Algorithm 1 Uncertainty-Aware Prior Router

```

1: Input: Hidden state  $h_\ell$ , Threshold  $\tau$ , Steering Vector  $v_{prior}$ 
2: Calculate Entropy  $H = -\sum p_i \log p_i$ 
3: Calculate Conflict Score  $C(\ell)$ 
4: if  $H > \text{Entropy\_Threshold}$  and  $C(\ell) > \tau$  then
5:   // High Uncertainty + High Prior Dominance detected
6:    $h'_\ell \leftarrow h_\ell - \alpha \cdot v_{prior}$ 
7:   Return  $h'_\ell$  (Steered State)
8: else
9:   Return  $h_\ell$  (Original State)
10: end if

```

This algorithm ensures we **do not steer** when the model is confident or when the image is consistent with priors (e.g., a yellow banana), preserving general capability. This is our key advantage over static methods like [11].

4 4. Execution Plan (8-Week Sprint)

Week	Focus Area	Deliverable
Week 1	Data Gen	Generating ‘CounterPrior-50’ via Diffusion.
Week 2	Baseline	Reproduce LLaVA failure (>30% error).
Week 3	Diagnosis	Plot ”Conflict Heatmaps” using Logit Lens.
Week 4	Vector Math	Extract steering vectors.
Week 5	Intervention	Implement dynamic steering code.
Week 6	Validation	Run POPE Benchmark [4] on A100.
Week 7-8	Writing	Submit to EMNLP.

5 5. Resource Strategy

To achieve statistical significance for the paper, I have structured my resource usage as follows:

- **Phase 1 (Dev):** Local Environment + Colab T4.
- **Phase 2 (Validation):** I will require access to the **Lab Compute (A100)** in Week 6 to run the full POPE benchmark [4].

6 6. Conclusion

This project targets a specific, high-value problem: **fixing hallucination without retraining**. By explicitly scoping this research to Static Image-Text interactions (while acknowledging the broader multimodal context), we ensure mechanistic depth and computational feasibility. The rigorous mathematical formulation of the "Routing Failure" provides a solid foundation for a top-tier publication.

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